Automatic Network Topology Generation & Validation Tool (Cisco VIP 2025)

Abstract

This project proposes a tool that can automatically generate a hierarchical network topology from router configuration files. It validates network configuration (duplicate IPs, wrong VLAN/gateway, MTU mismatches, loops), checks bandwidth and traffic capacity, recommends load-balancing or protocol changes (OSPF vs BGP), and supports Day-1/Day-2 simulations (ARP, neighbor discovery, OSPF discovery, link failures). The architecture is based on multithreading (routers/switches as threads) with IPC for communication and per-node logging.

Problem Statement

At present, there is no existing solution that can automatically create a network topology from router configuration files and simultaneously validate, optimize, and simulate the network. A tool is needed to achieve this efficiently, aligned with the concepts taught in advanced networking courses.

Objectives

- Automatically build network topology from configuration files.
- Check link bandwidth vs traffic demand.
- Detect configuration errors (duplicate IPs, wrong gateways, VLAN mismatches, MTU mismatches, loops).
- Suggest network optimization (node aggregation, protocol recommendation, load balancing).
- Perform Day-1 and Day-2 network simulations, including link failures.
- Implement using multithreading with IPC (FIFO/TCP) and per-node logging.

System Architecture

Modules:

1. **Input Parser** – Reads router configs and extracts IPs, VLANs, routing, MTU, bandwidth.
2. **Topology Builder** – Creates hierarchical graph of routers, switches, and endpoints.
3. **Validator** – Detects missing configs, duplicate IPs, wrong VLANs, wrong gateways, MTU mismatches.
4. **Traffic Model** – Assigns application-based load to endpoints.
5. **Optimizer** – Suggests load balancing, ECMP, protocol (OSPF/BGP) changes.
6. **Simulation Engine** – Day-1 discovery and link failure simulation.
7. **Reporting** – Summarizes issues, optimizations, and performance results.

Workflow

1. Load configuration files.
2. Parse device information (IPs, VLANs, interfaces).
3. Build topology graph (routers, switches, links).
4. Validate configuration (check for errors).
5. Apply traffic loads and check link utilization.
6. Suggest optimization (load balancing, protocol choice).
7. Run simulation (Day-1 boot and fault injection).
8. Generate final report with logs and recommendations.

Validation Rules

- Duplicate IP detection in same VLAN/subnet.
- Gateway address must match connected router interface.
- VLAN labels must match access/trunk configuration.
- MTU mismatch across a path must be flagged.
- Missing switch/router configs reported.
- Loops detected in L2/L3 topology.

Simulation Scenarios

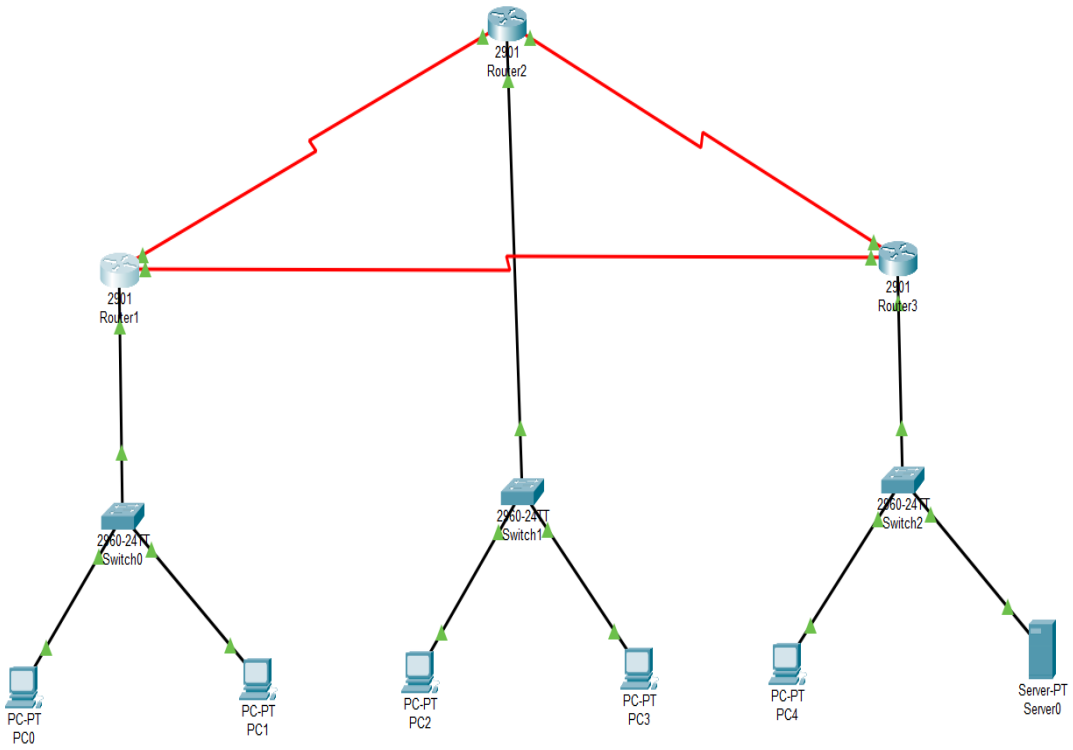
- **Day-1 Simulation:** ARP, OSPF neighbor discovery, initial ping tests.
- **Fault Injection:** Link down → rerouting and endpoint impact.
- **MTU Mismatch:** Detect and measure packet loss.
- **Pause/Resume:** Change configs and re-run simulation.

Implementation Architecture

- **Threads:** Each router/switch runs as a thread.
- **IPC:** FIFO or TCP channels for packet exchange.
- **Logs:** Each node keeps statistics (packets sent/received, errors).
- **Optional:** Real packet generation.

Packet Tracer Demonstration

- **Topology:** 3 Routers, 3 Switches, 5 PCs, 1 Server.
- **Connections:** Straight-through cables for LAN, Serial DCE for WAN.
- **IP Plan:**
    - R1 LAN – 192.168.10.0/24
    - R2 LAN – 192.168.20.0/24
    - R3 LAN – 192.168.30.0/24
    - WAN Links – 10.0.0.0/30, 10.0.0.4/30, 10.0.0.8/30
- **Routing:** OSPF configured on all routers.
- **Verification:** Successful ping tests between LANs and server.



Server0

Physical Config Services Desktop Programming Attributes

Command Prompt X

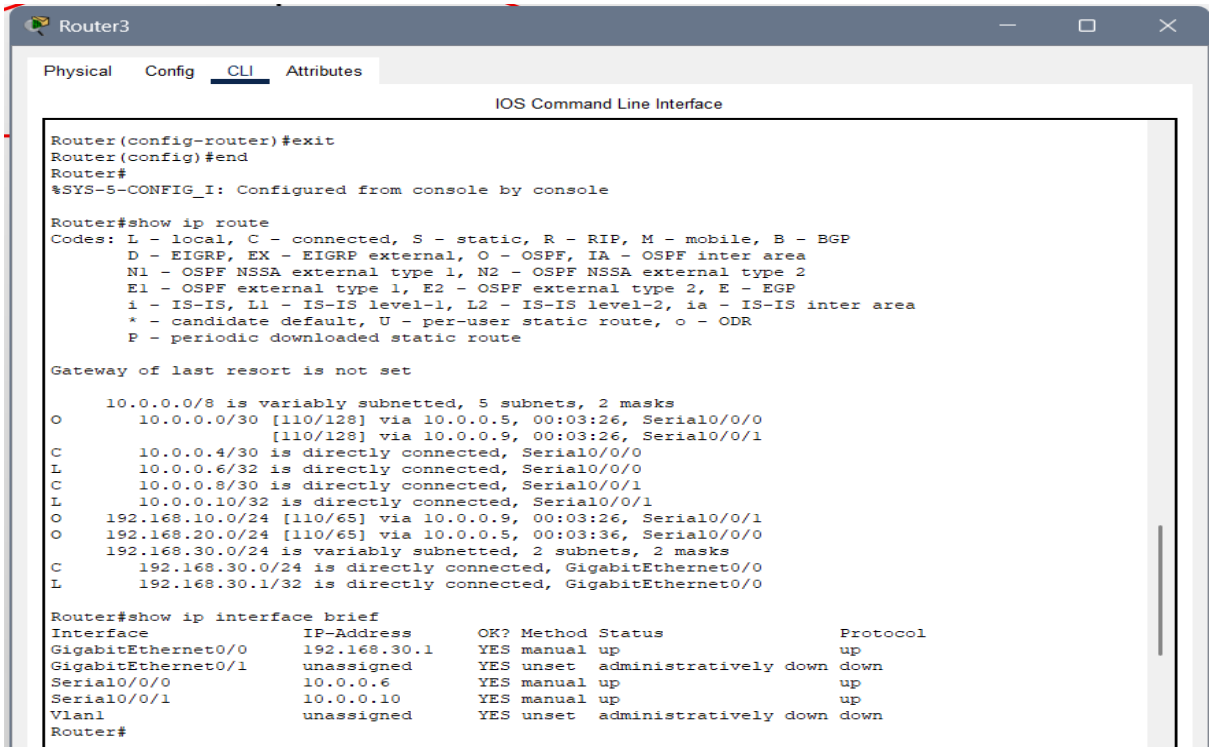
```
Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.20.2

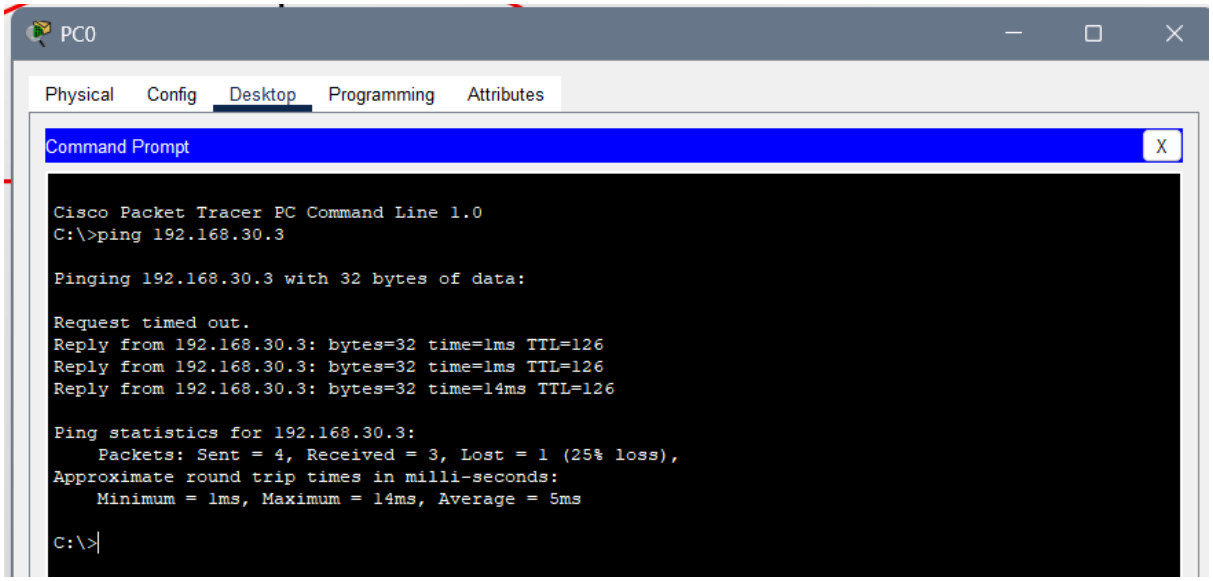
Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time=19ms TTL=126
Reply from 192.168.20.2: bytes=32 time=16ms TTL=126
Reply from 192.168.20.2: bytes=32 time=1ms TTL=126
Reply from 192.168.20.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 19ms, Average = 9ms

C:\>
```





Results

- Network topology generated successfully.
- OSPF routing worked with dynamic path selection.
- Errors like MTU mismatch and high utilization detected.
- Optimization suggestions given (load balancing and BGP recommendation).

Conclusion

- The tool design shows how to automate topology generation, configuration validation, and simulation. It improves troubleshooting, reduces manual effort, and ensures efficient and fault-tolerant networks. Future work includes GUI dashboards and machine learning-based traffic prediction.

Course: Networking Essentials – Cisco

Name – Sourabh Namdev

Enrolment No – LNCDBTC21122

Branch – CSE(AIML)

University – LNCT University